

Thermal, Viscosimetric and Thermomechanical Combined Assessment of Mixture Modelled Composite Fuels for Hybrid Propulsion

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Abstract: Paraffin-based fuels have received attention in hybrid propulsion studies towards their regression rate assessment, although little efforts have been dedicated to a consistent selection of their formulations. This work deals with melting enthalpy, dynamic viscosity, and linear thermal expansion characterization of novel low-density polyethylene-macrocrystalline paraffin-ammonium nitrate composite fuels, followed by model fitting under mixture

modeling principles and subsequent optimization through the desirability method. This approach allowed a better understanding of the raw materials' roles and provided a trade-off formulation, which was satisfactorily produced and characterized in a confirmatory experiment, thus making it a suitable candidate for further regression rate measurements.

Keywords: Hybrid propulsion · Polyethylene · Paraffin · Ammonium nitrate · Mixture modeled

1 Introduction

Hybrid propulsion has been recognized as a promising technology due to its unique combination of safety and simplicity, as compared to solid and liquid propulsions, respectively. Nevertheless, its performance remains a considerable challenge, with typical polymeric fuel regression rates being roughly an order of magnitude lower than usual solid propellants' burning rates [1,2]. In this sense, paraffin-based alternatives provide improvements in regression rates due to the entrainment effect, i.e., the mass transfer mechanism of molten fuel droplets into the oxidizer flow. This mechanism is driven by the shear stresses developed by the oxidizer flow over a low viscosity, hydrodynamically unstable layer of molten fuel [3,4].

In the development of hybrid propulsion, regression rate is in fact the ultimate performance criterium. Its results are largely dependent on the experimental settings, i.e., fuel and oxidizer nature, oxidizer flow rate and injector type, chamber geometry, fuel grain configuration, and fuel production method [1,2,5,6]. This abundance of factors can turn the interpretation of results from different studies into a difficult task [2,7]. Solid propulsion is not hampered by this disadvantage, because the burning rate of solid propellants can be taken, in most cases, from strand burner apparatuses, independently from the configuration of a rocket motor [1]. Coherently, the experimental assessment of regression rate as a dependent variable from hybrid motor operational settings, the building of dependable correlations for its prediction, and the attainment of robust and comprehensive mathematical models for combustion in hy-

brid systems have been considered as three of the most important challenges for hybrid propulsion in the 21st century [8,9].

Regarding both pure paraffins and thermoplastic polymers-paraffin composite fuels, literature has pointed out the performance relevance of fuel sample production assessment [10–12] and formulation selection based on physical properties [10, 13, 15]. Moreover, polyethylene-paraffin composite fuels have been shown to deliver both mechanical quality and combustion efficiency, while keeping appreciable gains in regression rates as compared to pure polymers [15–17].

In this scenario, increasing hybrid propulsion's technology readiness level might benefit from an initial commitment between solid fuel assessment under suitable analytical criteria and grain production feasibility at a lab scale. This last aspect has been recently investigated regarding a novel family of formulations comprising low-density poly-

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ethylene (LDPE), macrocrystalline paraffin and ammonium nitrate, which have been successfully prepared under a lab-scale centrifugal casting protocol [19].

Within these ternary formulations, it is possible to simultaneously investigate some performance-related parameters, i.e., melting enthalpy, dynamic viscosity and linear coefficient of thermal expansion, whose combined assessment can lead to the selection of a balanced fuel composition, before regression rate measurements. Hence, the approach adopted in this study aims at the rational choice of polyethylene-paraffin-based fuel formulations, considering some appropriate figures of merit. Accordingly, vaporization enthalpy has been recognized as being inversely proportional to the regression rate for purely polymeric fuels [20–22]. Regarding polyethylene-paraffin based fuels, chiefly characterized by melt-layer formation [3], it seems reasonable to state that fuel's melting enthalpy would play, to a considerable extent, the role of vaporization enthalpy in the abovementioned inverse relation, i.e., melt layer development would control regression rate [18,23,24]. Likewise, dynamic viscosity's inverse relation to regression rate has been pointed out for paraffin-based fuels [3,15,25,26] and the possibility of predicting regression rate through viscosity measurements has been raised as well [27,28]. By its turn, linear coefficient of thermal expansion, hereinafter designated as LCTE, may bring useful data regarding solid fuel dimensional stability under simultaneous thermal and mechanical loads common in rocket motor operation, given its role in restrained thermal expansion stresses and thermal shock resistance [29]. Moreover, thermal expansion understanding is important because of ammonium nitrate presence in the ternary fuel formulations, an oxidizer known for its polymorphic transitions and corresponding volumetric contractions/expansions [30].

Therefore, this work aimed at jointly assessing, through the desirability method [31], melting enthalpy, melt fuel dynamic viscosity and LCTE of novel low-density polyethylene-macrocrystalline paraffin-ammonium nitrate formulations [19]. This approach led to the selection of a suitable candidate fuel formulation for regression rate measurements, thus representing a trade-off among thermal, viscosimetric, and thermomechanical properties of paraffin-based fuels for hybrid propulsion. The selected formulation was successfully produced in a confirmatory experiment, according to the defined lab-scale protocol. Finally, a preliminary evaluation of the open burning behavior of the selected formulation in terms of repeated ignition-extinction capacity, a major feature of hybrid rocket motors [8,32], was conducted as well, with satisfactory results.

2 Experimental Section

The low-density polyethylene-macrocrystalline paraffin-ammonium nitrate composite fuel samples were produced ac-

cording to a lab-scale protocol previously developed, based on two major steps, i.e., mixing and centrifugal casting [19], which can be outlined as follows: (1) raw materials weighing and mold assembly; (2) LDPE and paraffin pre-mixing in the mold, under 120 rpm, at ambient temperature; (3) mold immersion in a glycerin bath at 150 °C, melting and mixing at 50 rpm until the bath temperature returns to the initial setting; (4) stirring at 100 rpm, with the bath temperature at 150 °C, for 15–25 minutes; (5) slow addition of $297 < \Phi < 590 \mu\text{m NH}_4\text{NO}_3$, under 100 rpm stirring and with the bath temperature at 150 °C; (6) homogenization for 15 min at 100 rpm, bath temperature kept constant at 150 °C; (7) removal of the stirrer, followed by the upper stopper and lid assembly, horizontal centrifugal casting at 400 rpm for 120 min and subsequent demolding of the composite fuel grain. Figure 1 shows some of these steps, identified by the numbers defined above. This process generates circular single-port fuel grains, suitable both for sampling for analytical purposes and static hybrid motor firings. Fuel formulations, shown in Table 1, stemmed from the extreme vertices technique, applied under mixture modeling principles [19,33].

Thermogravimetry (TGA) and Differential Scanning Calorimetry (DSC) measurements were carried out on a Netzsch STA449F3 simultaneous thermal analyzer, previously calibrated. The experimental conditions were set as follows: platinum crucibles, 20 mL/min N_2 protective atmosphere, 80 mL/min synthetic air purge atmosphere, 10 °C/min heating-cooling rate. The complete temperature program comprised an initial dynamic step (heating) from 30 °C to 140 °C, followed by a first isothermal step at 140 °C for 10 min, by an intermediate dynamic step (cooling) from 140 °C to 30 °C, by a second isothermal step at 30 °C for 50 min and by a final dynamic step (heating): from 30 °C to 600 °C. The



Figure 1. Some steps in the fuel samples' preparation method.

Table 1. Formulations of the composite fuels.

Order	Code	Original components proportions (wt%)		
		A (LDPE)	B (Paraffin)	C (NH ₄ NO ₃)
1	9C(2)	39.43	50.57	10.00
2	7C(1)	42.86	47.14	10.00
3	8C(1)	36.00	54.00	10.00
4	2V	47.62	52.38	0.00
5	4V	32.00	48.00	20.00
6	1V	40.00	60.00	0.00
7	9C(2)	39.43	50.57	10.00
8	6C(1)	35.05	44.95	20.00
9	5C(1)	43.81	56.19	0.00
10	9C(2)	39.43	50.57	10.00
11	3V	38.10	41.90	20.00

round-shaped samples, roughly 5.0 mm in diameter, were extracted from 1 mm thick discs, previously cut on a lathe from the middle longitudinal portion of each of the composite grains. The samples' mass ranged from 10 to 25 mg. Melting enthalpies were acquired from DSC traces, while TGA curves allowed the investigation of the confirmatory experiment's thermal decomposition pattern.

Dynamic viscosity assessment was provided by a Brookfield analogic viscometer model LVT. Number 3 spindle was selected considering the viscosities' range expected for the composites, i.e., 20 to 400,000 cP, which is comparable with viscosity data available for some low LDPE content-macrocrystalline paraffin composites [16,28]. Measurement conditions were set as a part of the ternary composites' preparation protocol [19], thus providing the experimenter with a preliminary, *in situ* estimate of the fuel viscosity. The idea was not to get an exact figure for viscosity but to collect comparative data in a predefined experimental region, useful to identify increase/decrease tendencies. Hence, they were adjusted from the standard Brookfield procedures, i.e., with the partial immersion of the spindle in the molten fuels and the direct use of the tabled Brookfield factors. Because of the non-Newtonian behavior of the fuels herein studied, different shear rates were required to allow stable measurements. Once homogenization was complete after the preparation step (6) described above, the stirrer was removed and the viscometer was brought into place, with the melt level slightly above the spindle's disc. The composite's temperature was set to 130 °C. Upon temperature stabilization and shear rate selection, dynamic viscosity readings were taken. Finally, the viscometer was withdrawn before the repetition of the final mixing cycle (step 6), followed by centrifugal casting (step 7).

Thermomechanical Analysis (TMA) conducted on a Netzsch DMA242E dynamic mechanical analyzer, previously calibrated, provided the LCTE data. The experimental conditions were set as follows: compressive mode, N₂ static atmosphere, cooling by liquid N₂, 10 °C/min cooling rate, and 5 °C/min heating rate. The temperature program comprised an initial dynamic step (cooling) from 20 °C to −100 °C, fol-

lowed by an isothermal step at −100 °C for 15 min and by a final dynamic step (heating): from −100 °C to 45 °C. The round-shaped samples, roughly 8.0 mm in diameter, were extracted from the middle radial portions of 5.5 mm thick discs, previously cut on a lathe from each of the composite grains. Samples' mass ranged from 250 to 290 mg.

The ignition-extinction assessment consisted in repeatedly exposing the fourth part of a 5.5 mm thick disc of the composite sample, under ambient pressure, to an oxygen-rich oxyacetylene flame for nearly 2 s. The result was considered positive if extinction occurred in less than 10 s after removal of the torch, as recorded by a model EOS 60 D Canon camera.

Interpretation of the results was conducted under mixture modeling principles [33], using MINITAB19 software.

3 Results and Discussion

Table 2 summarizes the composites' results, organized in the random preparation order of the samples. The next subsections bring the discussions and interpretations re-

Table 2. Thermal, viscosimetric and thermomechanical results of the composite fuels.

Order	Code	Melting enthalpy (J/g)	Dynamic viscosity (cP)	Linear coefficient of thermal expansion ($\times 10^{-6}/^{\circ}\text{C}$)
1	9C(2)	230.4	14,840.0	98.73
2	7C(1)	227.0	31,266.7	71.27
3	8C(1)	234.1	8,120.0	87.15
4	2V	240.7	16,886.7	85.20
5	4V	229.8	9,023.3	0.46
6	1V	228.2	6,916.7	106.56
7	9C(2)	221.2	9,593.3	54.81
8	6C(1)	204.6	18,740.0	−4.38
9	5C(1)	227.9	11,166.7	96.85
10	9C(2)	225.1	11,586.7	92.81
11	3V	202.3	28,666.7	−5.52

garding the data obtained under the selected analytical criteria.

3.1 Thermal Analysis – Melting Enthalpy

Melting enthalpy was acquired in the DSC second heating step, in the 30 °C to 120 °C approximate interval, which was selected to allow the inclusion of complete LDPE and paraffin melting and thus fully encompass melt layer formation and subsequent entrainment mechanism [4] activation. This temperature range does not include ammonium nitrate fusion (circa 170 °C). The samples were cut from the middle radial portion of 1 mm thick discs. The results are displayed in Table 2. Figure 2 depicts DSC traces in the melting interval for (a) three composite formulations and their raw materials and (b) for the same composites alone, i.e. Nouv_2 V_00, Nouv_3 V_00, and Nouv_7 C(1)_00 corresponding to points 2 V, 3 V and 7 C(1) in Tab. 1, respectively. These formulations share the same B/A ratio, i.e., 1.1, which yielded excellent macroscopic homogeneity and structural integrity [19]. In addition, they range from ammonium nitrate absence (Nouv_2 V_00) to its maximum amount (Nouv_3 V_00), i.e., 20 wt%.

Figure 2 (b) shows two major melting peaks, corresponding to paraffin (circa 65 °C) and LDPE (circa 97 °C), considerably overlapped as compared to the pure raw materials. The resulting temperature gap of nearly 32 °C is considerably smaller than the one from pure organic components under the same experimental conditions (circa 41 °C), as can be seen by comparing Figure 2 (a) and (b). This suggests an acceptable LDPE-paraffin miscibility level allowed by the preparation protocol, in the same pattern previously observed for crystallization enthalpy [19]. The attained re-

duction of the melting temperature gap, regardless of the solid oxidizer amount, would benefit the entrainment occurrence in a hybrid motor, given the appropriate oxidizer mass flux is provided. Apart from differences regarding raw materials' specifications, formulation range, and experimental settings, this microstructural interaction level was not attained in previous studies, possibly due to mild preparation conditions [17]. In addition, Figure 2 (a) also reveals an endothermic phase transition for NH_4NO_3 circa 56 °C, most probably corresponding to the volumetric expansion expected to occur at nearly 32.3 °C, which in this case may have been delayed due to experimental factors [34].

Melting enthalpy results, ranging approximately from 200 to 240 J/g, are expressively lower than ablation enthalpy for polyethylenes, i.e., circa 3,556 J/g [35]. They were fit to a reduced quadratic model, shown in Eq. 1, alongside with the standard errors for the coefficients, where "h fus" stands for melting enthalpy. ANOVA for this model indicated that the regression is significant, and lack of fit is not, whereas R^2 and the maximum explainable variability amounted to 89.7% and 96.8%, respectively, thus suggesting an acceptable quality fit [33,36]. There are two interaction terms, the first between LDPE (x_1) and paraffin (x_2), and the second between LDPE (x_1) and ammonium nitrate (x_3), both strongly decreasing melting enthalpy, a quality shared by the single oxidizer coefficient, but to a much lesser extent. Accordingly, the other individual coefficients' magnitudes suggest a smaller role of isolated organic materials in increasing the response.

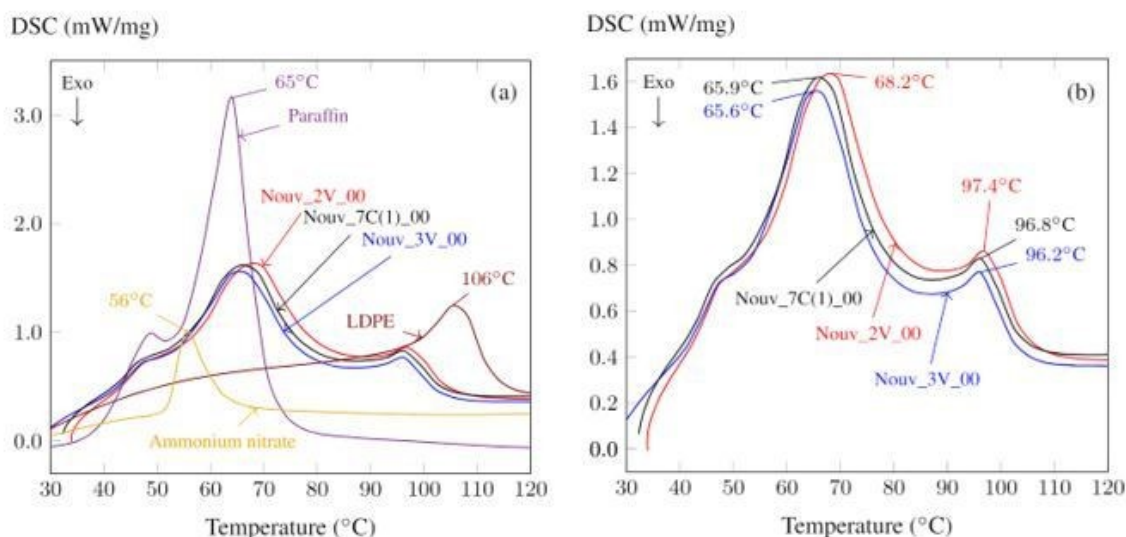


Figure 2. DSC traces in the melting interval for (a) three composite formulations and their raw materials and (b) for the composites alone.

$$\begin{aligned} \hat{y}_{\text{hfus}} = & \underset{(595)}{2,973} x_1 + \underset{(338)}{1,801} x_2 - \underset{(232)}{313} x_3 \\ & - \underset{(1,834)}{8,486} x_1 x_2 - \underset{(758)}{3,476} x_1 x_3 \end{aligned} \quad (1)$$

These melting enthalpy interactions in the ternary LDPE-paraffin-ammonium nitrate composites are illustrated by Figure 3, which shows (a) the contour plot and (b) the Cox response traces for the fitted quadratic model (Eq. 1). In Figure 3 (a), the experimental points are labeled according to Table 1, as well as the grey lines define the borders of the experimental region. Cox traces allow a direct interpretation of each of the raw materials' effects on the response, provided small deviations from a hypothetical reference blend are implemented [33].

These pictures reveal the strong dependence of melting enthalpy on LDPE and paraffin amounts. NH_4NO_3 , on the other hand, does not melt in the 30 °C to 120 °C range. Nevertheless, the magnitude of its individual coefficient in Eq. 1 and its trace in Figure 3 (b) suggest that increasing amounts of the solid oxidizer would slightly reduce melting enthalpy. This would be beneficial under the fuel melt layer formation viewpoint [18,23] and would possibly counter-balance an increase in dynamic viscosity due to ammonium nitrate loading, as discussed in the next section. In this context, other aspects such as processability and hygroscopicity must be considered as well. Moreover, both LDPE and paraffin traces in Figure 3 (b) clearly show that melting enthalpy reaches a minimum in the vicinities of the reference blend. This formulation, depicted in Figure 3 (b), although not prepared, is quite close to the centroid, point 9C(2) in Tab. 1, thus suggesting that an ideal composite, regarding solely melting enthalpy, would lie in the centroid

region. Nonetheless, the multivariate approach adopted in this study, stemming from the very nature of the problem, requires viscosimetric and thermomechanical aspects to be considered as well.

3.2 Viscosity Assessment

Dynamic viscosity data obtained during the preparation of the composite grains are presented in Table 2. The attained range at 130 °C, from circa 7,000 to 32,000 cP, has shown itself compatible with the devised preparation method, but it is quite expressive as compared to pure paraffin. The macrocrystalline paraffin used in this work yielded a dynamic viscosity of 8.7 cP under the measurement settings established, a figure comparable with literature data [27]. On the other hand, the above-mentioned viscosity range is expressively lower than the one estimated for pure LDPE, i.e., 500,000 cP at 180 °C [37]. This confirms the intermediate behavior expected for the assessed composites in terms of viscosity. These data were fit to a reduced linear model, presented in Eq. 2, along with the standard errors for the coefficients. ANOVA for this linear model suggested that regression is significative, and lack of fit is not, whereas R^2 and the maximum explainable variability amounted to 77.1 % and 97.9 %, respectively, thus suggesting, as expected, a poor fit [33,36].

$$\hat{y}_{\text{Dynvisc130}^\circ\text{C}} = \underset{(29,347)}{151,613} x_1 - \underset{(22,923)}{97,584} x_2 + \underset{(16,190)}{47,354} x_3 \quad (2)$$

The quality of the fitted model stems from the *in-situ* measurement conditions, which included shear-thinning effects [38]. Accordingly, there were no interaction terms, de-

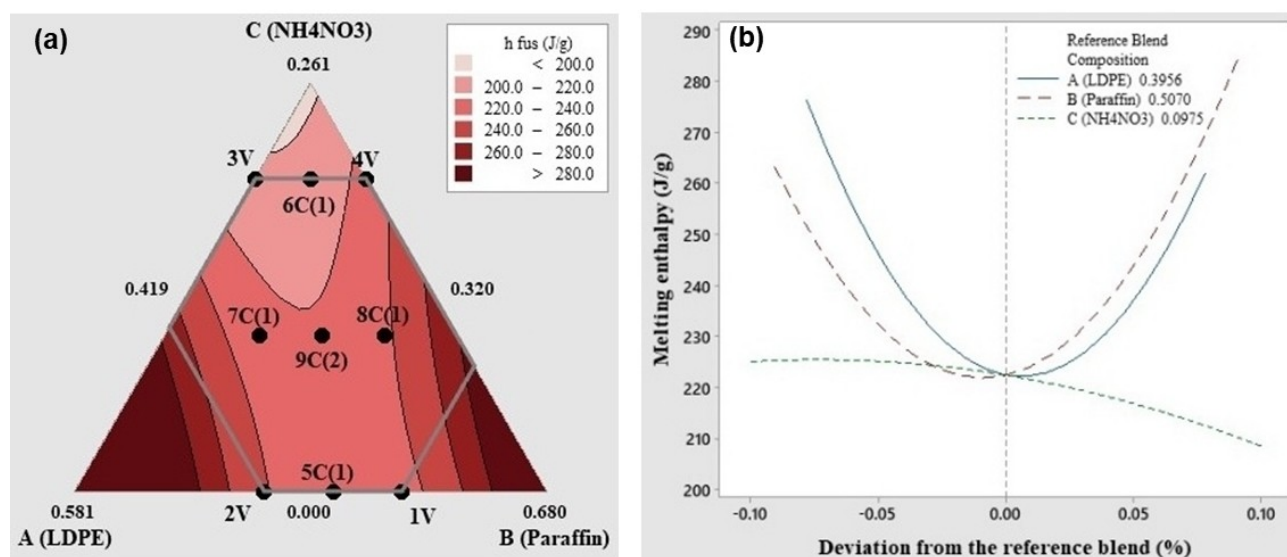


Figure 3. Melting enthalpy (a) contour plot and (b) Cox traces.

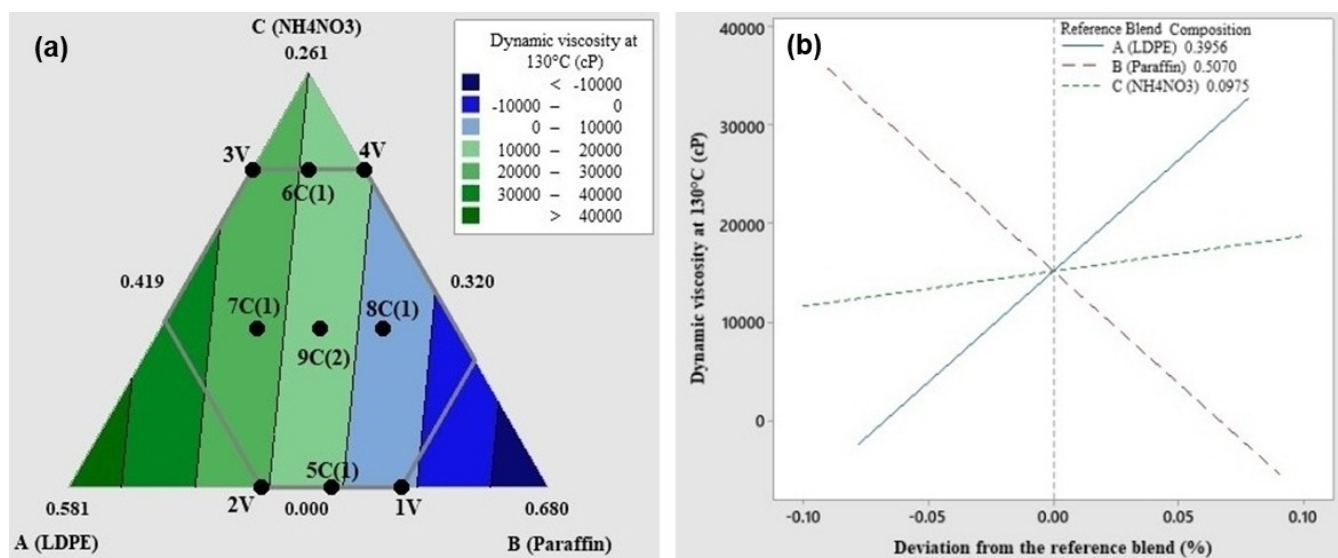


Figure 4. Dynamic viscosity (a) contour plot and (b) Cox traces.

spite the considerable shear stresses involved in the mixing step of the preparation method and the resulting partial miscibility level observed through melting and crystallization enthalpies [19]. Nonetheless, given the focus of these preliminary measurements not on accuracy but the relative behavior of the formulations, the attained results allowed some useful interpretations. These can be seen in Figure 4, which shows (a) the contour plot and (b) the Cox response traces for the fitted linear model (Eq. 2).

These plots show that, as observed during the composites' preparation, increments in LDPE and NH₄NO₃ proportions increase viscosity, the contrary being true for paraffin. The magnitudes of the coefficients in Eq. 2 suggest that, as expected, LDPE exerts the main effect. Cox response traces also show the relevant role played by paraffin in decreasing viscosity. The positive solid oxidizer effect on viscosity, although much less pronounced, reinforces the correctness of reducing the upper NH₄NO₃ limit to control hygroscopicity and improve processability [19]. Finally, it sounds reasonable to hypothesize that the viscosity increment stemming from the need to produce feasible composite grains [19] could perhaps be counterbalanced by gains in regression rates and by a higher combustion efficiency [23]. These are linked to the possible effect of NH₄NO₃ burning mechanism, primarily characterized by the initial melting of the oxidizer [39], in this case, embedded in the LDPE-paraffin matrix.

3.3 Thermomechanical Analysis

The linear deformation traces corresponding to the same composite fuel samples depicted in Figure 2 (b) are shown in Figure 5. As before, these formulations share a 1.1 B/A

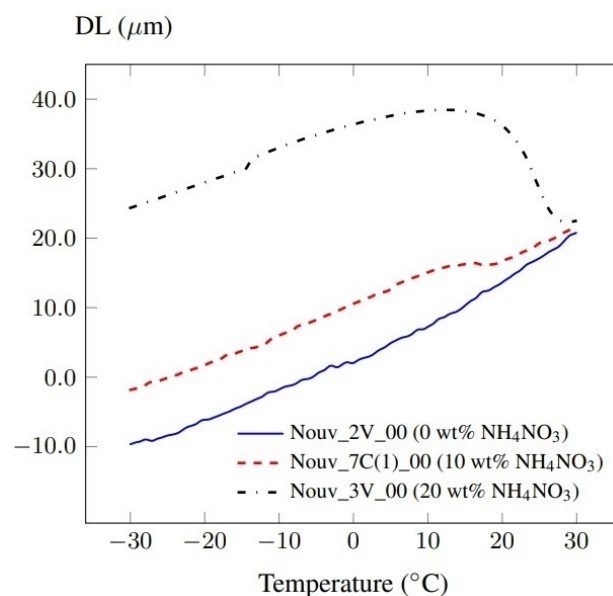


Figure 5. Linear deformation traces for Nouv_2 V_00, Nouv_7 C(1)_00 and Nouv_3 V_00 composites.

ratio and vary from the absence of solid oxidizer, a 10 wt% NH₄NO₃ amount, and a 20 wt% NH₄NO₃ content. The selected temperature range, from -30 °C to 30 °C, was intended to encompass a wide assessment perspective, but not to allow paraffin melting.

In Figure 5, one can notice the quasi-linear expansion trace corresponding to the formulation without NH₄NO₃, as compared to the expansion curve containing a slight plateau-like region around 10 °C for the 10 wt% oxidizer formulation, and finally the quite distinct expansion-contraction profile displayed by the composite containing

20 wt% oxidizer. Accordingly, their thermal expansion coefficients amounted to $85.02 \times 10^{-6}/^{\circ}\text{C}$, $71.27 \times 10^{-6}/^{\circ}\text{C}$ and $-5.52 \times 10^{-6}/^{\circ}\text{C}$, respectively. These trends can be attributed to the combination of an increasing oxidizer content and its unique polymorphic change in the -100 to 30°C range, namely at -16.8°C , a 2.9% volumetric contraction [34,40]. This was not apparent at the proper temperature region probably due to the interaction of two factors, i.e., heating rate and samples' dimensions. As for the first factor, it was set at $5^{\circ}\text{C}/\text{min}$ [17], a figure considered as the upper rate limit to allow a proper thermal equilibrium within the sample [41]. Regarding samples' dimensions, despite complying with ASTM recommendations [42], they may have favored the development of thermal gradients. These experimental factors have probably enhanced the samples' thermal lag [43], therefore considerably delaying NH_4NO_3 contraction.

Visual inspection of the samples immediately after the runs did not reveal any external cracks. LCTE results, calculated [29] from traces like the ones in Figure 5, are depicted in Table 2. These data, ranging from -5.52 to $106.56 \times 10^{-6}/^{\circ}\text{C}$, were fit to a reduced quadratic model, shown in Eq. 3, alongside with the standard errors for the coefficients. ANOVA for this model indicated that regression is significant, and lack of fit is not. R^2 and the maximum explainable variability amounted to 94.1% and 94.2%, respectively, thus suggesting an excellent fit [33,36]. There are two interaction terms, the first between LDPE (x_1) and ammonium nitrate (x_3), and the second between paraffin (x_2) and ammonium nitrate (x_3), both strongly increasing thermal expansion. On the contrary, NH_4NO_3 exerts a remarkable individual effect in decreasing thermal expansion. The other individual coefficients' magnitudes suggest a much smaller role of isolated organic raw materials in increasing (paraffin) or decreasing (LDPE) the response.

$$\begin{aligned} \hat{y}_{\text{LCTE}-30\text{to}30^{\circ}\text{C}} = & \frac{68}{(135)} x_1 + \frac{224}{(105)} x_2 - \frac{3,155}{(677)} x_3 \\ & + \frac{3,937}{(1,421)} x_1 x_3 + \frac{3,057}{(1,226)} x_2 x_3 \end{aligned} \quad (3)$$

The thermal expansion interactions in the ternary LDPE-paraffin-ammonium nitrate composites can be further understood in Figure 6, which shows (a) the contour plot and (b) the Cox response traces for the fitted quadratic model (Eq. 3).

These graphics suggest that LCTE in the -30 to 30°C range is strongly affected by NH_4NO_3 and paraffin amounts, while LDPE seems to play a secondary role. Pure paraffin yielded a $98.45 \times 10^{-6}/^{\circ}\text{C}$ LCTE, thus comparable to most of the results in Table 2, while low-density polyethylenes' room temperature LCTE lie within the $180\text{--}400 \times 10^{-6}/^{\circ}\text{C}$ range [29]. Hence, it seems reasonable to state that paraffin controls the thermal expansion of the composites, in agreement with previous studies [17], except for the 20 wt% oxidizer content formulations. For these, NH_4NO_3 plays a major role, as can be seen by its distinct trace in Figure 6 (b). Moreover, in Table 2 it is noticeable that all the composites containing 20 wt% ammonium nitrate have yielded either contraction (3 V and 6 C(1)) or negligible expansions (4 V). This behavior stems from the abovementioned NH_4NO_3 polymorphic transition [34,40], which was probably delayed not only due to the adopted experimental conditions but also as a consequence of the solid oxidizer interactions with LDPE and paraffin, as evidenced in Eq. 3. Hence, the need to define an appropriate upper limit for NH_4NO_3 in the composites due to hygroscopicity and processability concerns is strengthened by the requirement of structural integrity under thermal cycling. In this sense, the 20 wt% figure herein

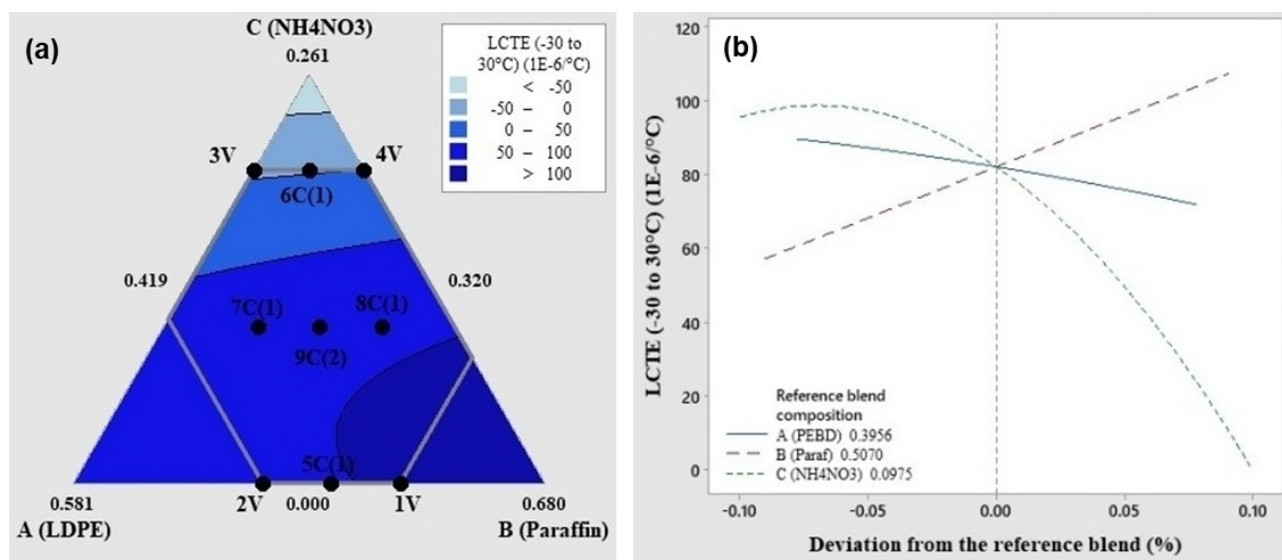


Figure 6. LCTE (a) contour plot and (b) Cox traces.

defined [19] seems adequate, with an ideal value probably lying circa 10 wt% NH_4NO_3 .

From a hybrid propulsion standpoint, the attained LCTE range, lower than some common polymeric fuels such as polyethylenes and poly (methyl methacrylate), seems adequate for its favorable implications in keeping composite fuel's dimensional stability, in preventing restrained thermal expansion stresses, and in increasing thermal shock resistance [1,29]. Thus, within the expansion range, which excludes the 20 wt% solid oxidizer composites, meeting these criteria would lead to LCTE minimization.

3.4 Combined Assessment through Desirability Method

Melting enthalpy, dynamic viscosity, and thermal expansion results can be jointly assessed to define a trade-off fuel formulation, capable of simultaneously meeting thermal, viscosimetric, and thermomechanical requirements of polyethylene/paraffin-based composite fuels for hybrid propulsion. Additionally, considering the importance of B/A ratio for structural integrity [19], it was included as a fourth parameter, directly linked to the composites' grains lab-scale manufacturing feasibility. The desirability method was selected to perform the requested multiobjective optimization because of its suitability to the problem under scrutiny [31,36]. As for the ranges adopted for desirability optimization, they were defined within the experimental intervals depicted in Fig. 3, 4, and 6. Regarding the previous results and discussions, all the parameters were set to be minimized. Hence, the dynamic viscosity interval was set to be 14,000–22,000 cP, LCTE range was set to be 70–100 $\times 10^{-6}/^\circ\text{C}$, and melting enthalpy should lie in the 220–230 J/g range. B/A ratio was set to be in the 1.10–1.29 range, because of previous structural integrity/macroscopic homogeneity results [19]. Desirability functions for all the parameters were given a 1.0 weight. The resulting optimized formulation consisted of 39.52 wt% LDPE, 47.95 wt% paraffin, and 12.53 wt% NH_4NO_3 . The individual desirabilities amounted to 0.3668 for dynamic viscosity, to 1.000 for

LCTE, to 0.8890 for melting enthalpy, and 0.4072 for B/A ratio, so that combined desirability [31] amounted to 0.6037, reflecting not only the abovementioned requirements but also the quality of the fitted models.

Subsequently, the optimized formulation was prepared in a confirmatory experiment according to the lab-scale protocol [19] and analyzed in terms of melting enthalpy, dynamic viscosity, and thermal expansion under the aforementioned experimental settings. Figure 7 shows the corresponding composite grain (a) top view; (b) side view and (c) 1 mm thick disc cut from its middle portion. It was labeled Exp_confirm_00.

It yielded a 3.55 cm length, a 6.51 cm and 2.69 cm external and internal diameters, respectively, as well as a volumetric contraction of 9.58%, comparable to that of pure paraffin prepared under the same method, i.e. 9.90% [19]. In addition, its mass amounted to 89.29 g, thus allowing a 9.73% theoretical porosity, also comparable to the 11.33% figure for pure paraffin, both attained in a centrifugal casting process, without vacuum application, and thus significantly higher than the ones found in other studies [44,45]. This result seems coherent with the decrease in volumetric contraction stemming from centrifugal casting in pure paraffin, as compared to static casting data [19]. Nonetheless, Figure 7 (a) and (b) show that its dimensional regularity is noticeable, particularly across the port, as well as within the composite grain, as can be seen in Figure 7 (c), whose absence of major cracks and voids suggests that the attained porosity is related to minor bubbles of entrapped air, evenly distributed across the composite grain. As for macroscopic homogeneity, the apparent concentration of solid oxidizer in the peripheric region, seen in Figure 7 (c), can be better understood considering Figure 8, which shows the TGA (a) and the DSC (b) traces of the samples collected at the outer, middle and inner positions of a given radial segment of the disc in Figure 7 (c). All thermal curves refer to the last dynamic heating step (30 $^\circ\text{C}$ –600 $^\circ\text{C}$).

Both TGA and DSC traces suggest reasonably close profiles for melting and thermal decomposition stages, the first one up to circa 200 $^\circ\text{C}$. In Figure 8 (a), the trace for the outer

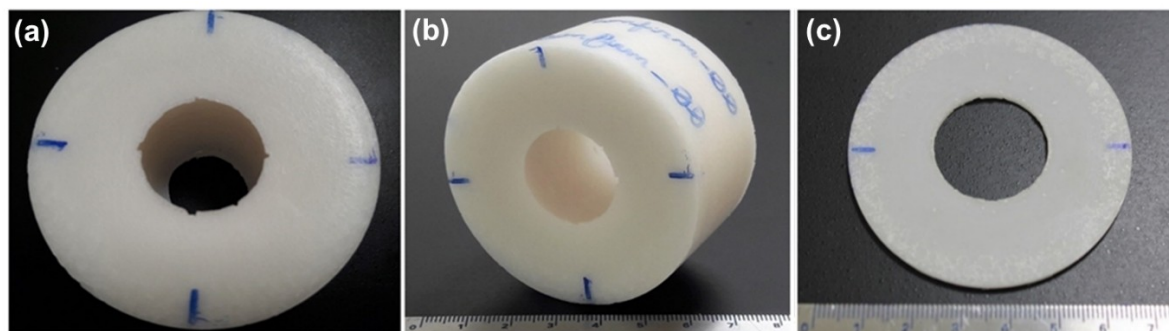


Figure 7. Composite grain corresponding to the confirmatory experiment (a) Top view; (b) Side view and (c) 1 mm thick disc cut from its middle portion.

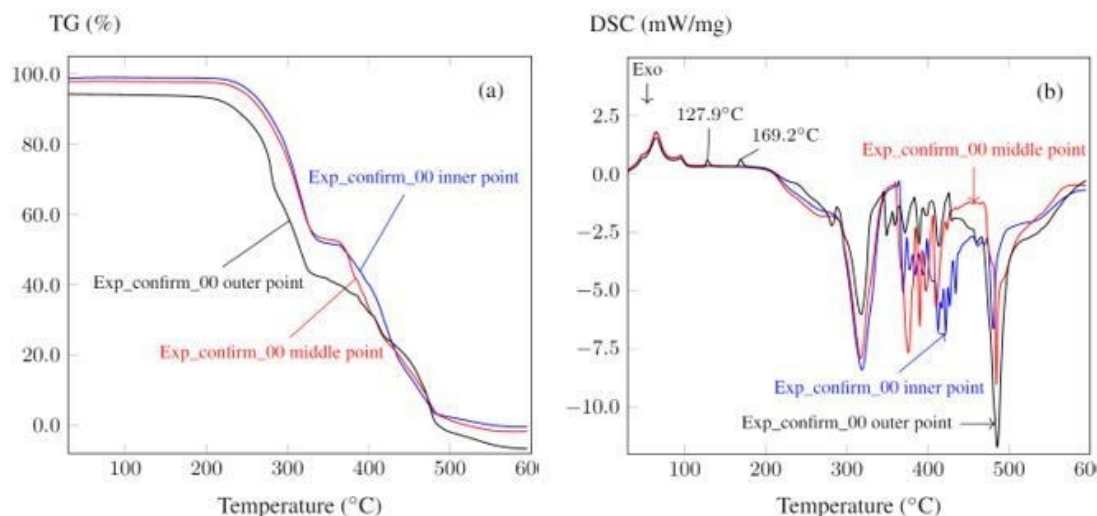


Figure 8. (a) TGA and (b) DSC traces of samples collected from a given radial segment of composite disc Exp_confirm_00.

position exhibits a higher mass loss from the start of the final dynamic step (circa 30 °C), probably due to the higher concentration of NH_4NO_3 on this region, whose moisture was lost during the first dynamic heating up to 140 °C. This evidence of moisture loss in the peripheric region agrees with the apparent NH_4NO_3 accumulation, depicted in Figure 7 (c). Accordingly, in Figure 8 (b) the only noticeable difference among the three positions in the melting stage (up to circa 200 °C) lies on the peaks associated with a solid-solid transition of NH_4NO_3 (127.9 °C) and with its melting (169.2 °C) on the outer position curve. As for thermal decomposition, both TGA and DSC traces suggest its occurrence in three consecutive stages, among which the second one is the less distinct. TGA traces indicate that the mass losses for the outer, middle, and inner positions in the 30 °C–600 °C range were 99.91 %, 99.57 %, and 99.31 %, respectively, amounting to an average mass loss of 99.60 % and a relative standard deviation [14] of 0.30 %. These results indicate that, under a partially oxidative atmosphere, the attained NH_4NO_3 radial dispersion is compatible with a uniform mass loss behavior along the burning direction, i.e., across the radial axis. Furthermore, from a rocket motor performance standpoint, unburnt fuel fraction minimization is of paramount importance [1]. Table 3 presents some other data extracted from Figure 8.

Applying Dixon criterion [14] to the detection of outliers to each data subset in Table 3 reveals that, except for onset temperature at the inner position, none of the extreme results (suspects) differs significantly from the respective data subset, at the 95 % confidence level. The inner onset temperature increment suggests a delay in its thermal decomposition start as compared to the middle and outer positions. This behavior agrees with the macroscopic assessment provided by Figure 7 (c), i.e., a relative decrease in solid oxidizer amount in the inner region may have

Table 3. Quantitative data for samples collected from a given radial segment of composite disc Exp_confirm_00.

Sample	Temperature (°C)			Melting enthalpy (J/g)
	Onset	Offset	Melting peak	
Exp_confirm_00 outer position	274.4	482.0	64.4	194.2
Exp_confirm_00 middle position	274.3	489.4	64.6	220.3
Exp_confirm_00 inner position	278.1	480.5	64.6	217.0

slightly delayed its thermal decomposition. In short, Exp_confirm_00 has satisfactorily attained structural integrity criterion and partially met macroscopic homogeneity requirement, although NH_4NO_3 radial dispersion does not seem to appreciably affect the thermal behavior along the burning direction.

Melting enthalpy, dynamic viscosity, and LCTE for the optimized formulation were assessed according to the described experimental procedures. The results are summarized in Table 4, together with their model predictions pro-

Table 4. Theoretical and experimental figures for the trade-off formulation.

Parameter	Model prediction	Experimental value	Relative error (%)
Dynamic viscosity at 130 °C (cP)	19,065.3	16,533.3	–13.28
LCTE –30 to 30 °C ($\times 10^{-6}/^\circ\text{C}$)	64.0	73.2	14.38
Melting enthalpy (J/g)	221.1	220.3	–0.36

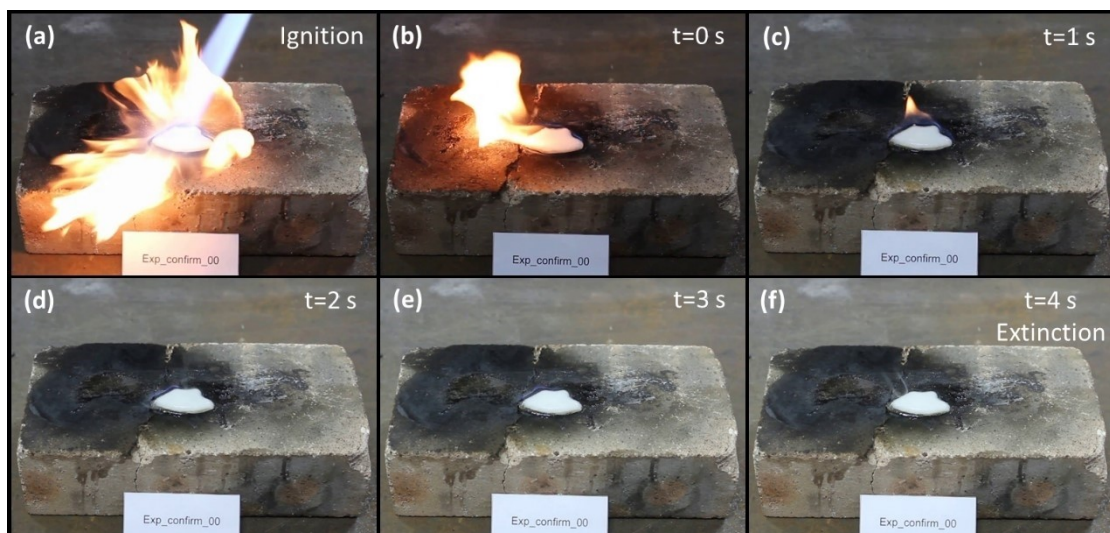


Figure 9. Exp_confirm_00 ignition (a) and subsequent extinction (b–f) at ambient pressure.

vided by Eq. 1, 2, and 3, respectively. The corresponding relative errors [14] are displayed as well, taking the theoretical predictions as references. As for viscosity, the well-known limitations of its experimental procedure can explain the considerable relative error. On the other hand, LCTE yielded an unexpectedly high relative error, given its excellent model fitting. This could be explained by TMA thermal lag and by a slightly lower NH_4NO_3 content in the sample, associated with the abovementioned radial dispersion, thus favoring an expansion behavior.

Finally, a preliminary open burning was conducted with Exp_confirm_00 sample to assess its ignition-extinction capacity. This is an important performance requirement of hybrid rocket motors, to which NH_4NO_3 presence in the composite fuel can be harmful [8]. Figure 9 shows (a) the third consecutive successful ignition and (b–f) the corresponding complete extinction in less than 4 s of the optimized formulation, in the same pattern observed with the pure paraffin sample tested under these conditions.

This result confirms the suitability of the trade-off formulation, containing 12.53 wt% solid oxidizer, to the repeated ignition-extinction criterium at ambient pressure. As stated before, this is a preliminary evaluation that must be confirmed by hybrid motor static firings, in which the development of chamber pressures shall reveal the ignition-extinction behavior under real operational conditions. In any case, if the trade-off formulation had failed this ambient pressure-screening test, serious concerns could be raised towards its applicability as fuel for hybrid propulsion.

4 Conclusion

Thermal, viscosimetric, and thermomechanical characterizations of low-density polyethylene-macrocrystalline paraf-

fin-ammonium nitrate composites were successfully accomplished, allowing acceptable model fittings under the mixture modeling principles and providing some useful hints on their behavior under the hybrid propulsion standpoint. Accordingly, solid oxidizer addition to the LDPE-paraffin organic matrix should be limited to 20 wt% not only by processability and hygroscopicity concerns but also by its effects in increasing dynamic viscosity and strongly decreasing thermal expansion to the verge of a minor contraction, thus hampering the fuel dimensional stability. In the experimental range herein investigated, this last parameter seems to be controlled by paraffin, the higher crystallinity raw material, except for the 20 wt% ammonium nitrate compositions. Melting enthalpy is markedly decreased, as expected, by LDPE inclusion, while NH_4NO_3 role is less pronounced. Desirability combined assessment of the experimental results led to an optimized 39.52 wt% LDPE/47.95 wt% paraffin/12.53 wt% NH_4NO_3 formulation, which was successfully fabricated as a confirmatory experiment under the lab-scale protocol previously defined, yielding a composite grain free from major structural defects, whose oxidizer radial dispersion did not appreciably affect its thermal behavior. Preliminary ambient pressure repeated ignition-extinction evaluation was successful as well, thereby making the attained trade-off ternary formulation a consistent candidate for further regression rate assessments in hybrid rocket motor/combustion chamber firings.

Symbols and Abbreviations

- A LDPE label in mixture modeling
- B Macrocrystalline paraffin label in mixture modeling
- C Ammonium nitrate label in mixture modeling
- x Components proportions (wt %)

- 1 LDPE subscript in components proportions
- 2 Paraffin subscript in components proportions
- 3 Ammonium nitrate subscript in components proportions

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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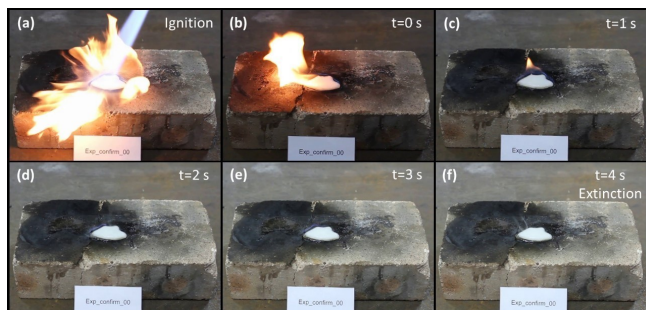
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RESEARCH ARTICLE



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**Thermal, Viscosimetric and Ther-
momechanical Combined Assess-
ment of Mixture Modelled
Composite Fuels for Hybrid Pro-
pulsion**
